

Dx4x2i Drilling adjustments



WARNING



IGNORING INSTRUCTIONS HAZARD!

To avoid death or injury you **MUST** read, understand and follow operator's and maintenance manuals before installing, inspecting, operating, servicing, testing, cleaning, transporting, storing, dismantling or disposing of the product or a part or accessory of the product. Keep this publication for future reference.

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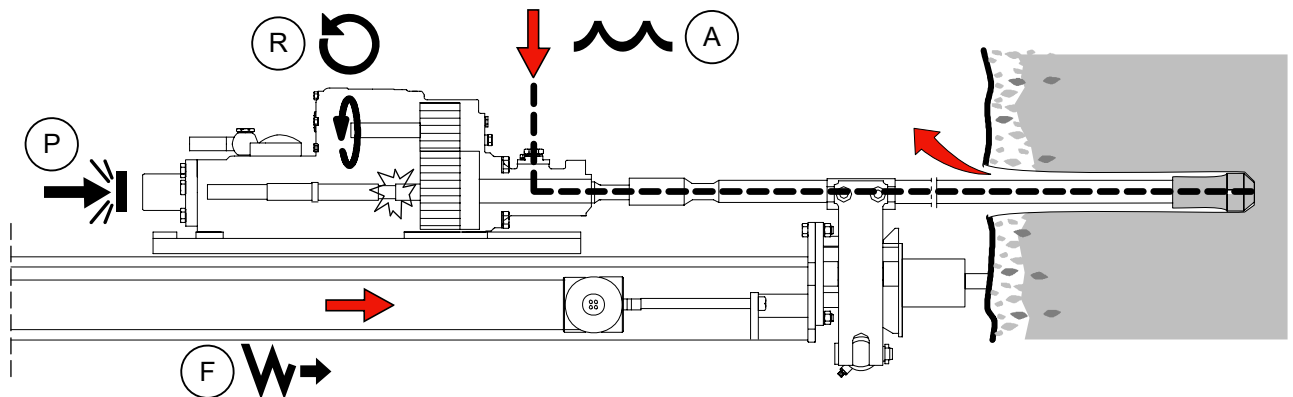
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1 ROCK DRILLING PROCEDURE AND EQUIPMENT

1.1 Principles of percussion drilling



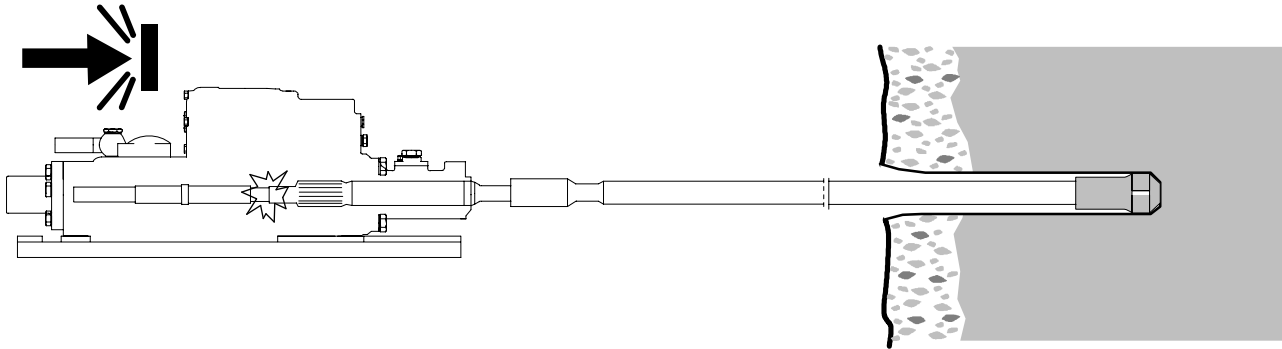
Percussion drilling has four main functions:

- Percussion (P), which makes the drill bit penetrate the rock.
- Feed (F), which keeps the drill bit in tight contact with the rock.
- Rotation (R), which turns the drill bit to a new position before the next impact.
- Flushing (A), which removes the cuttings and cools the drilling equipment.

The percussion energy created by the rock drill piston is transferred to the rock through the drilling tools.

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1.1.1 Percussion (P)



Percussion refers to the rock drill piston's impacting, reciprocating movement generated by the hydraulics. The rock drill's shank transfers the piston's percussion energy into the drill rod, and the rod does the same through the drill bit to the rock. The striking point is the point where the piston hits the shank.

The percussion power is generated by the percussion force, the percussion frequency, and the drill bit's penetration. The percussion power is conveyed to the rock through the drilling equipment. The percussion power can be directly controlled by using percussion pressure.

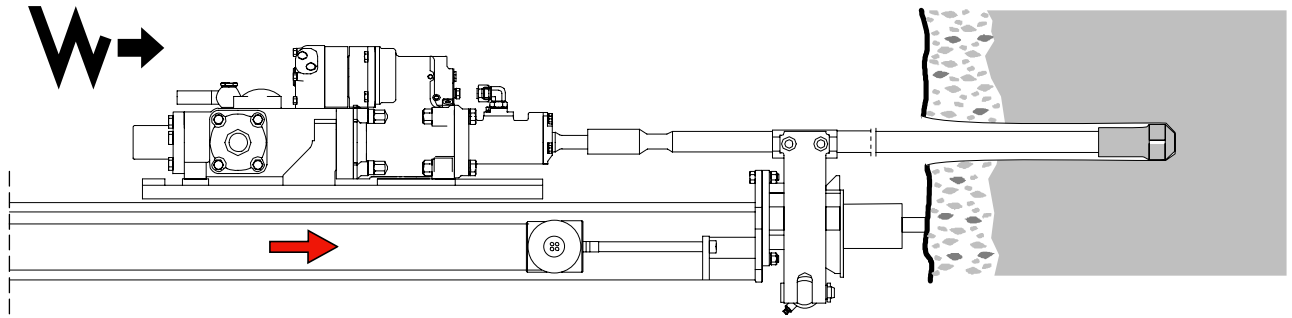
One of the basic issues in drilling is the energy transfer capacity of the drilling equipment. Only a certain maximum amount of kinetic energy can be transferred using drilling equipment of a certain size. When the equipment's ability to transfer energy is exceeded, equipment damage increases rapidly.

The percussion power used depends on the quality of the rock to be drilled. If the rock is soft, a lower percussion pressure can be used. If the rock is hard, a higher percussion pressure is used.

Setting the percussion pressure always involves a compromise between penetration rate and equipment durability.

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1.1.2 Feed (F)



The feed movement means the rock drill's linear movement on the feed.

The purpose of the feed is to keep the shank tightly connected to the rock drill and the drill bit in firm contact with the rock.

When the percussion pressure is increased, the feed pressure must also be increased. The correct feed power depends on the percussion pressure, rock conditions, hole depth, and the size and type of the drilling equipment.

Broken rock should be drilled using low percussion pressure and low feed pressure.

The correct feed pressure can be determined by observing and listening.

OBSERVE the following, which should be present:

- Constant progressive movement of the rock drill (the drill carriage should not shake)
- Constant rotation of the rock drill (constant rpm)
- Tight shank connection (the connection must not overheat / turn blue / emit smoke)
- Constant penetration

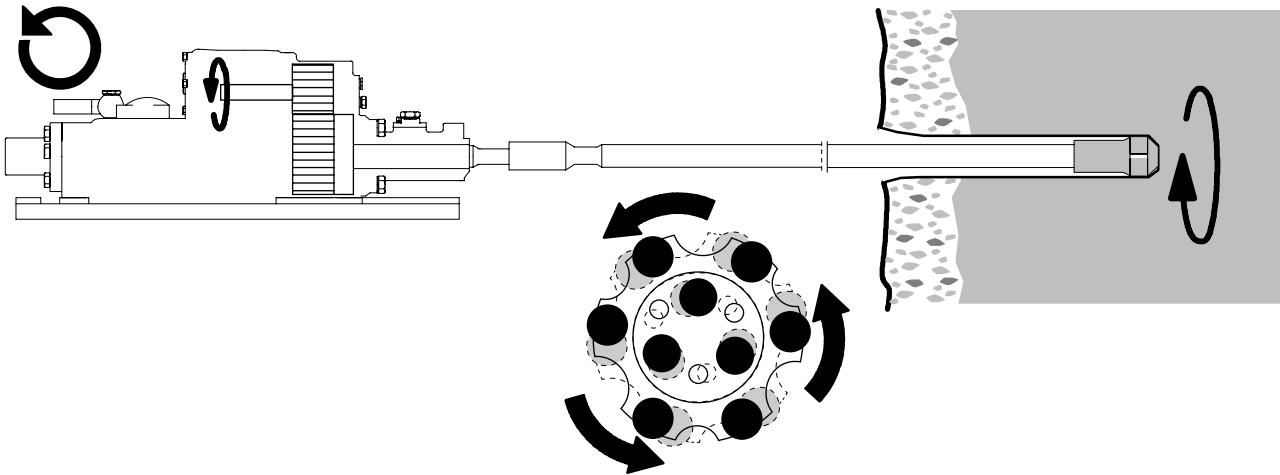
LISTEN for the following:

- Constant drilling sound (no rattling or shaking sounds)

Excessive feed power does not increase penetration. It only increases drill bit wear, causes irregularity in the drilled holes, and bends the rod, causing unusually high rotation pressures.

Inadequate feed power leads to loss of contact between the drill bit and the rock. This causes the percussion energy to be reflected back to the drilling equipment, causing damage to the equipment, the rock drill, and the drill carriage.

1.1.3 Rotation (R)



Rotation is the rotational motion of the drill rod that advances the drill bit buttons to a new position between percussion impacts. The hydraulic motor inside the rock drill rotates the drill rod through a gearing.

The correct rotation speed depends on the form and diameter of the drill bit, percussion pressure (frequency), hole depth, rock conditions, and the other main functions of drilling.

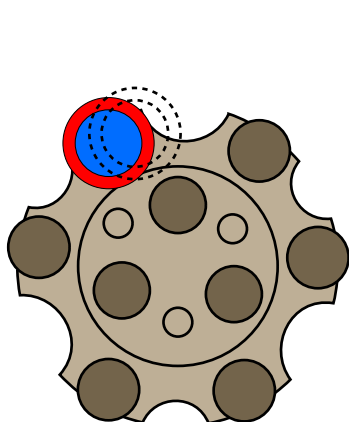
Inadequate rotation speed causes energy loss (the cuttings are ground too finely) and leads to poor penetration.

Too high a rotation speed results in excessive bit wear, since the rock is broken by rotation rather than percussion. Excessive rotation speed also leads to over-tightening of threaded connections.

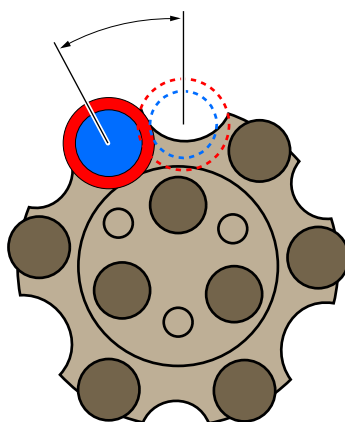
Dx4x2i Drilling adjustments

Rotation speed

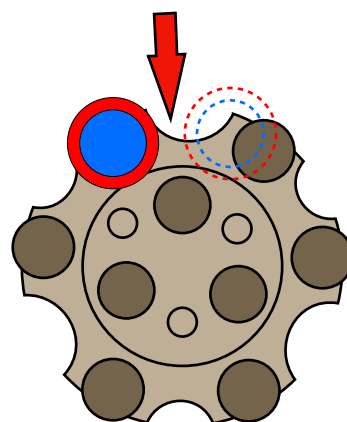
Rotation is used to create an indexing distance between two percussion impacts. By using the correct rotation speed, buttons wear from the middle.



Rotation speed too slow



Target rotation speed



Rotation speed too high

Too slow rotation speed will break the rock into very small pieces, like sand.

Too high rotation speed will break the rock into small stones.

The optimal theoretical rotation speed is based on percussion frequency and bit size.

$$\text{Rotation speed (rpm)} = \frac{60 \times f \times b}{\pi \times D}$$

f = percussion frequency (Hz)

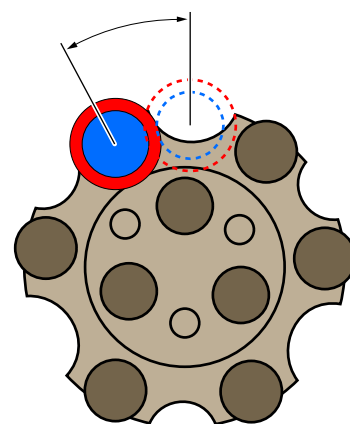
D = bit diameter (mm)

b = indexing distance (mm), can vary for different rock conditions

Example:

- RD525 93 Hz (235 bar)
- Bit diameter 48 mm
- Indexing distance 7 mm

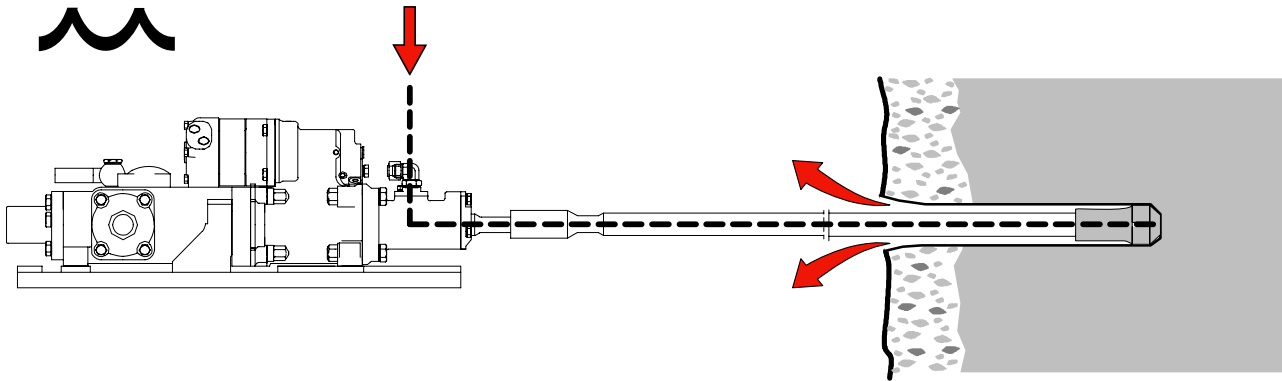
Target rotation speed = 259 rpm.



Indexing distance (b)

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1.1.4 Flushing (A)



Flushing means transferring the loose rock material (cuttings) created during drilling away from the drill hole. The other task of the flushing function is to cool the structures of the rock drill and the drilling equipment.

Poor flushing leads to low penetration (cuttings are ground at the bottom of the hole), short drilling equipment service life (the risk of rod jamming is increased), and rapid wear of the drill bit.

Water flushing

Water flushing is a flushing method that binds rock dust. Flushing is led to the rock drill's flushing housing, from which it flows through the shank to the drill rod, and onward through the flushing holes in the drill bit to the drill hole.

Air flushing

The flushing air is carried to the bottom of the hole through the rod hole and the holes in the drill bit. The mix of flushing air and cuttings is directed out of the hole through the space between the rod and the hole wall.

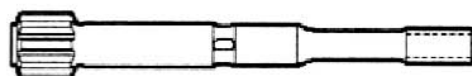
Air-mist flushing (optional)

In air-mist flushing, a certain amount of water is sprayed into the flushing air. The amount of water is adjusted according to the hole size, drilling direction, and rock type.

1.2 Drilling tools

Shank

The impact energy and rotation torque of a rock drill are transferred to drilling tools through the shank.



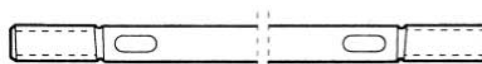
Coupling sleeve

Coupling sleeves connect extension rods to one another.



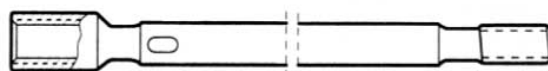
Extension rods

Both ends of an extension rod have identical threads. Rod length is usually 10...18 ft (300...540 cm).



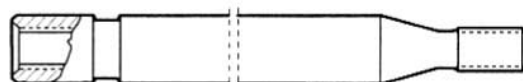
MF rods

MF rods have a male thread in one end and a female thread in the other. Rod lengths are the same as for extension rods.



Guide tubes

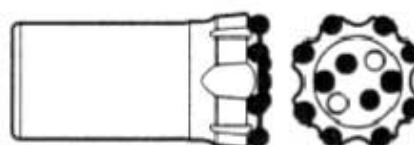
Guide tubes minimize hole deviations.



Drill bits

A button bit is the most common drill bit type. Its advantages are good penetration rate, durability, and easy re-sharpening.

Various types of button bits are available for different drilling conditions.



Dx4x2i Drilling adjustments

Example of the use of different drilling tools in a drill string

A guide tube and a special bit are used when straight holes are required and the drilling conditions are difficult.



- 1 Shank
- 2 Coupling sleeve
- 3 Rod
- 4 Drill bit

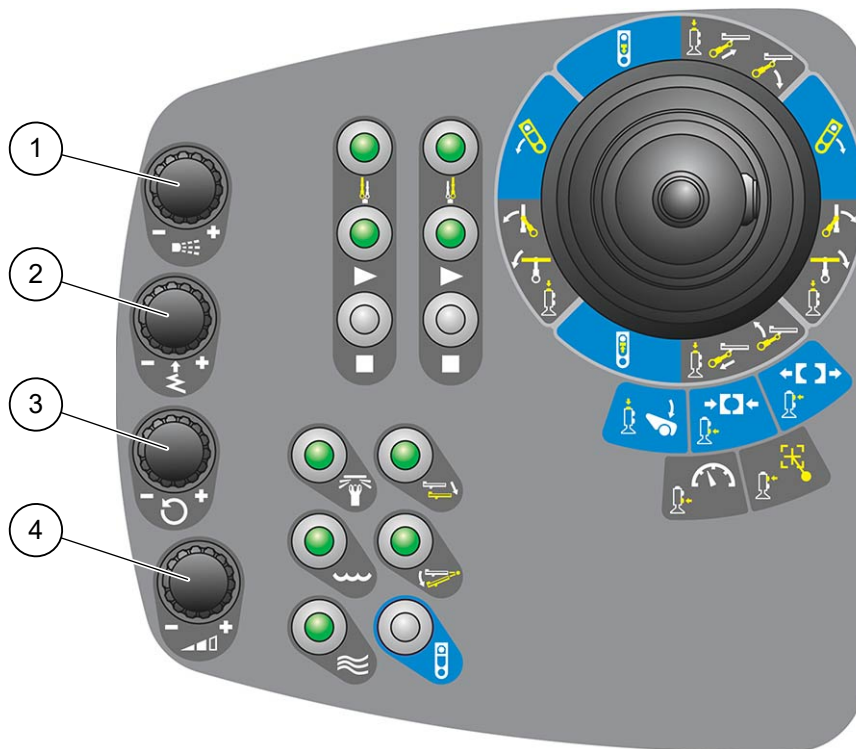


- 1 Shank
- 2 MF rod
- 3 Drill bit

2 CONTROLS

2.1 Drilling panels

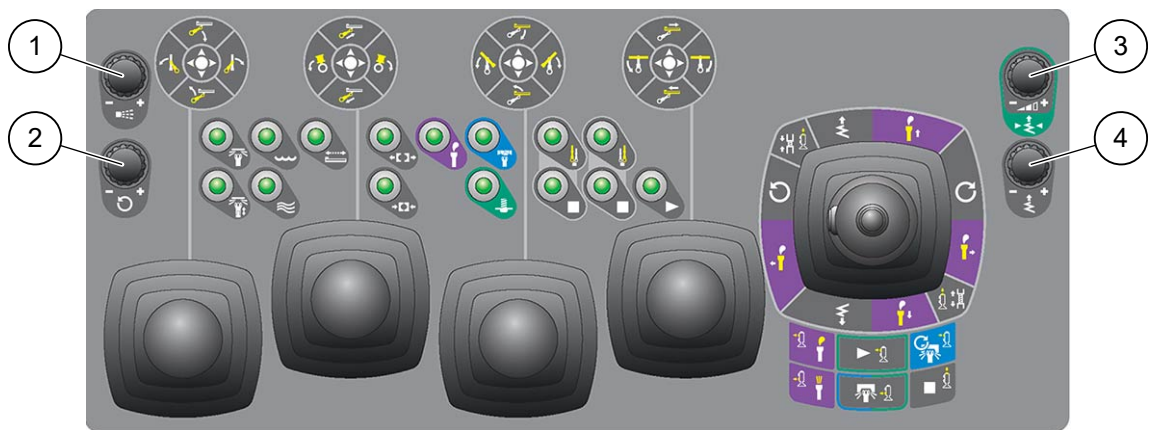
DD422i(E) Gold/Platinum



- 1 Adjusting knob for flushing
- 2 Adjusting knob for collaring speed / feed pressure
- 3 Adjusting knob for rotation pressure / rotation speed
- 4 Adjusting knob for drilling power

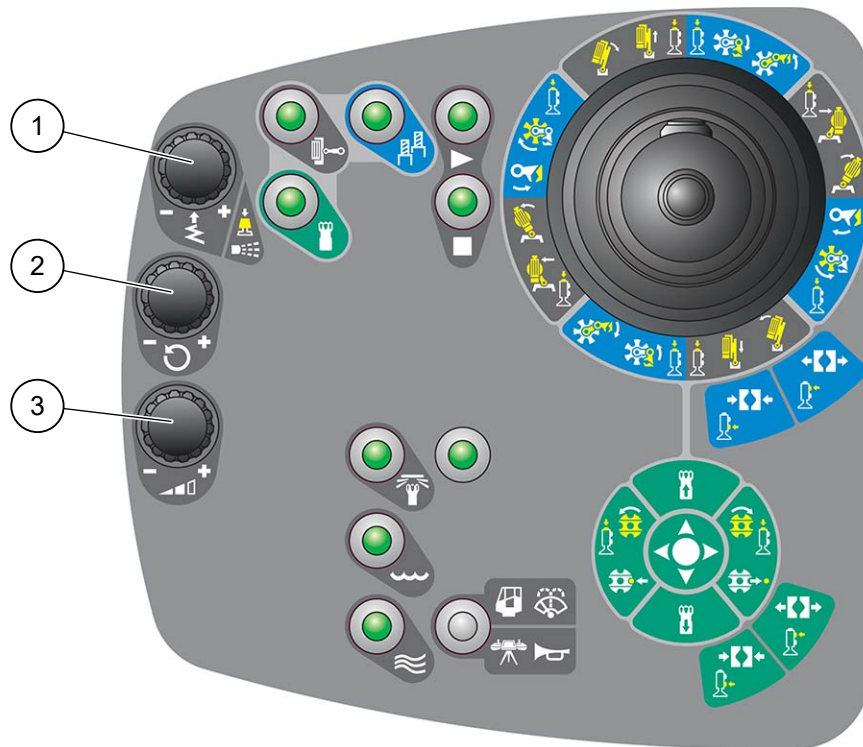
Dx4x2i Drilling adjustments

DD422i Dual Controls



- 1 Adjusting knob for flushing
- 2 Adjusting knob for rotation pressure / rotation speed
- 3 Adjusting knob for drilling power
- 4 Adjusting knob for collaring speed / feed pressure

DL4x2i(E) Gold/Platinum

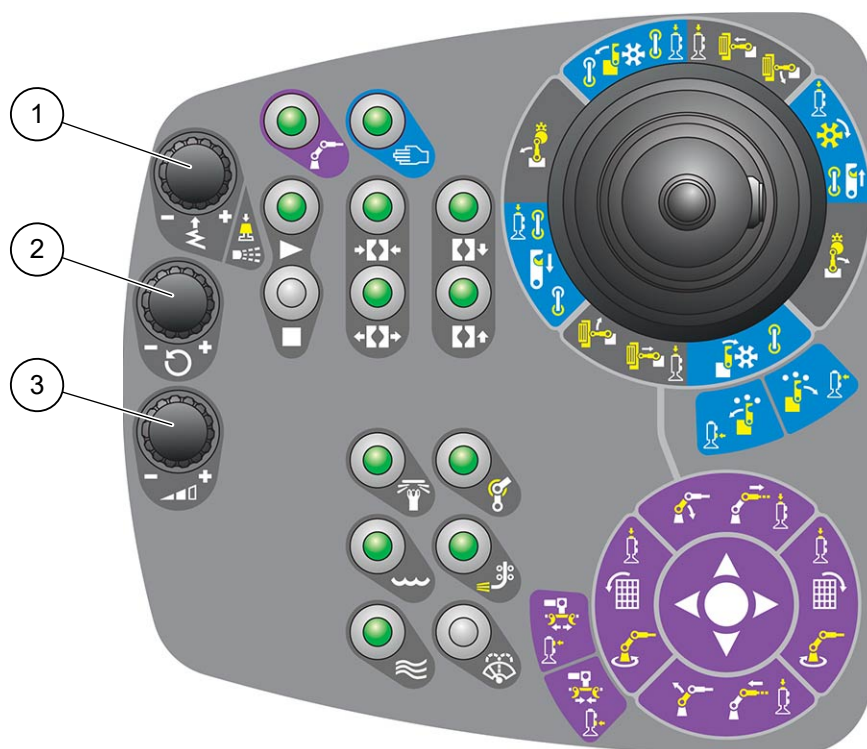


- 1 Adjusting knob for collaring speed / feed pressure, adjusting air-mist flushing (optional) when the knob is pushed
- 2 Adjusting knob for rotation pressure / rotation speed ¹
- 3 Adjusting knob for drilling power

¹ Rotation speed when the knob is pushed once and then rotated (in torque control mode and advanced torque control mode).

Dx4x2i Drilling adjustments

DS412i(E)

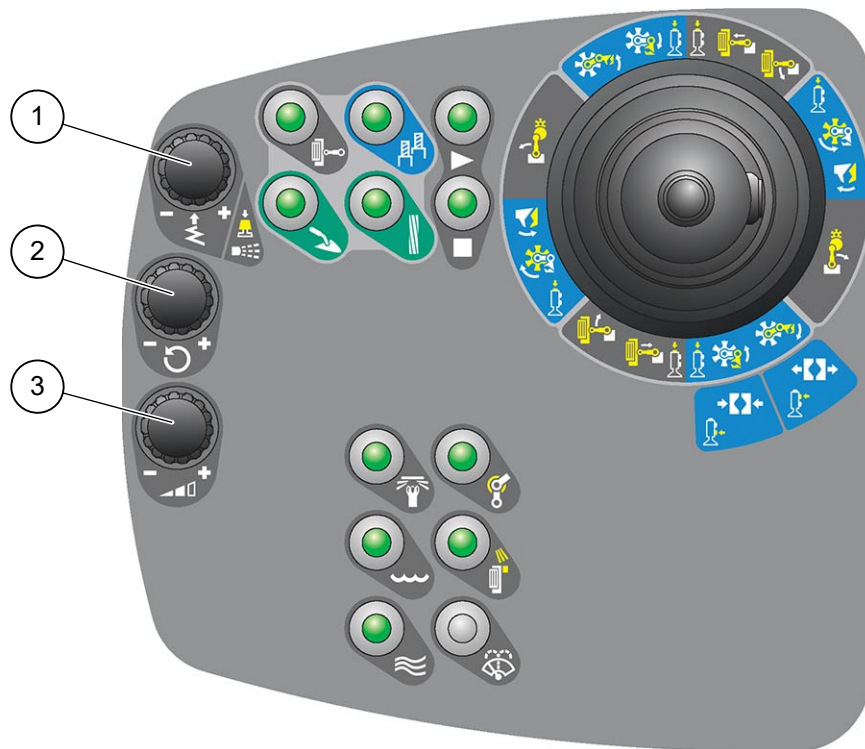


- 1 Adjusting knob for collaring speed / feed pressure, adjusting air-mist flushing (optional) when the knob is pushed
- 2 Adjusting knob for rotation pressure / rotation speed ¹
- 3 Adjusting knob for drilling power

¹ Rotation speed when the knob is pushed once and then rotated (in torque control mode and advanced torque control mode).

Dx4x2i Drilling adjustments

DS422i(E)



- 1 Adjusting knob for collaring speed / feed pressure, adjusting air-mist flushing (optional) when the knob is pushed
- 2 Adjusting knob for rotation pressure / rotation speed ¹
- 3 Adjusting knob for drilling power

¹ Rotation speed when the knob is pushed once and then rotated (in torque control mode and advanced torque control mode).

3 DRILLING ADJUSTMENT

3.1 Preconditions

Before adjusting any drilling settings, make sure that any existing machine condition will not affect the settings. Making adjustments under false conditions will result in poor settings, which will cause poor machine performance and premature wear of the equipment.

Note that some of the settings are general drilling settings, and some are recipe-specific settings.

Mechanical condition of the feed

- Rock drill must be able to move freely.
 - Ensure that the slide pieces of the rock drill carriage are correctly adjusted. Refer to the manuals of the machine in question.
- Feed must be correctly adjusted.
 - Refer to the manuals of the machine in question.

Rock tools

- Minimum requirement: a new drill bit
- Recommendation: new or very slightly worn drill steels.

Control system

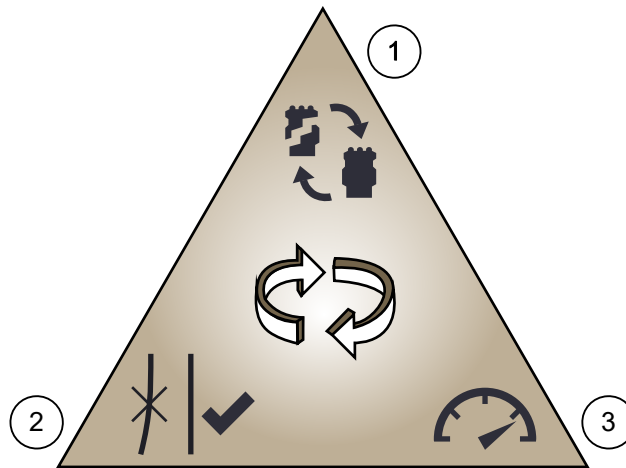
- Make sure that any power reductions are not affecting drilling.

Flushing

- Make sure that water and/or air flow and pressure provided from the mine's infrastructure are sufficient.
- Make sure that the water line from the mine's network to the machine is at least the same size as the water hose on the machine.

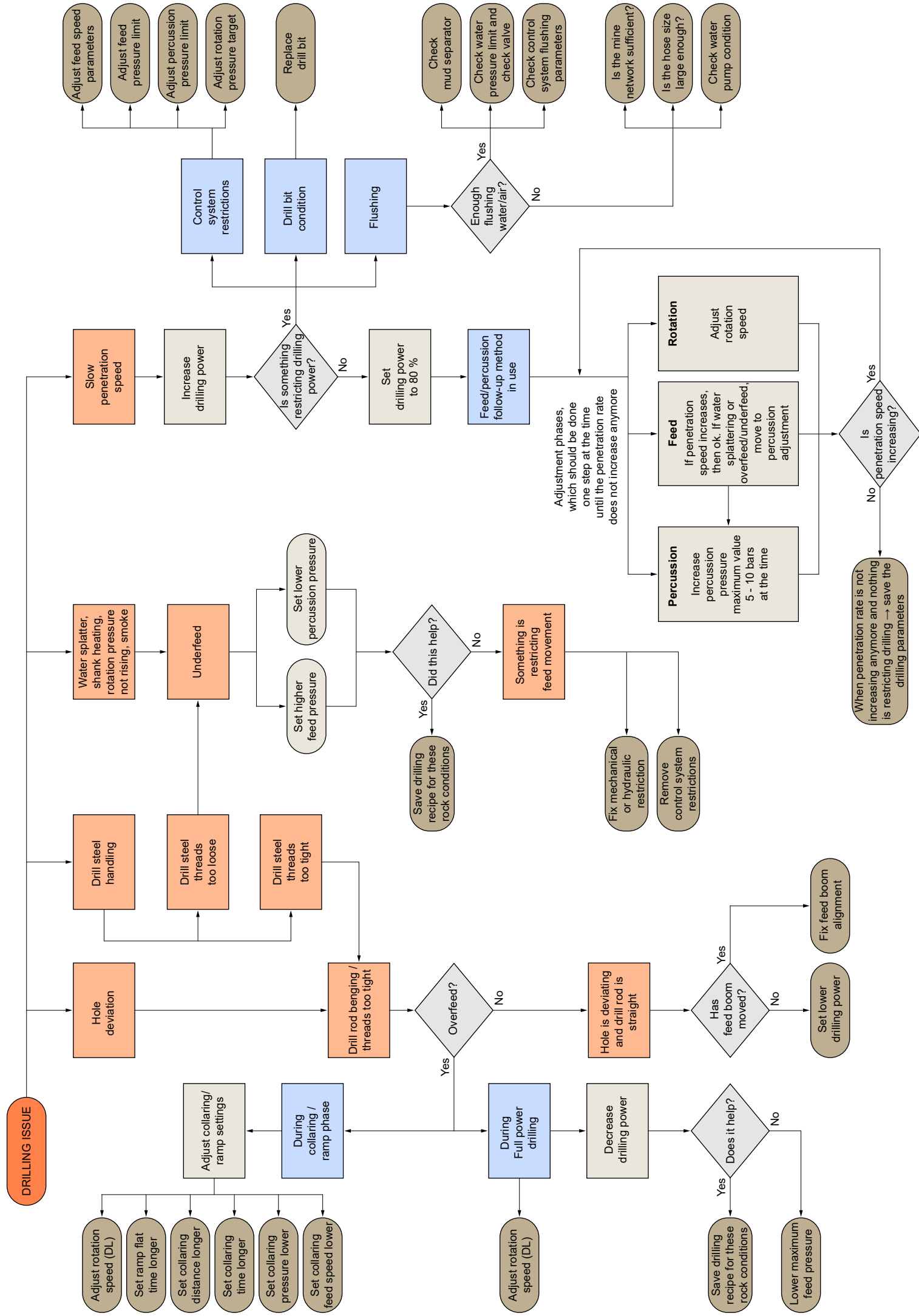
3.2 Drilling adjustment flowchart

The drilling adjustment flowchart is designed to provide a simplified drilling adjustment guide when some common drilling issues are encountered. The flowchart is not a comprehensive guide that suits every possible situation. Make sure that you take the machine, environment, and rock conditions into account when making any adjustments.

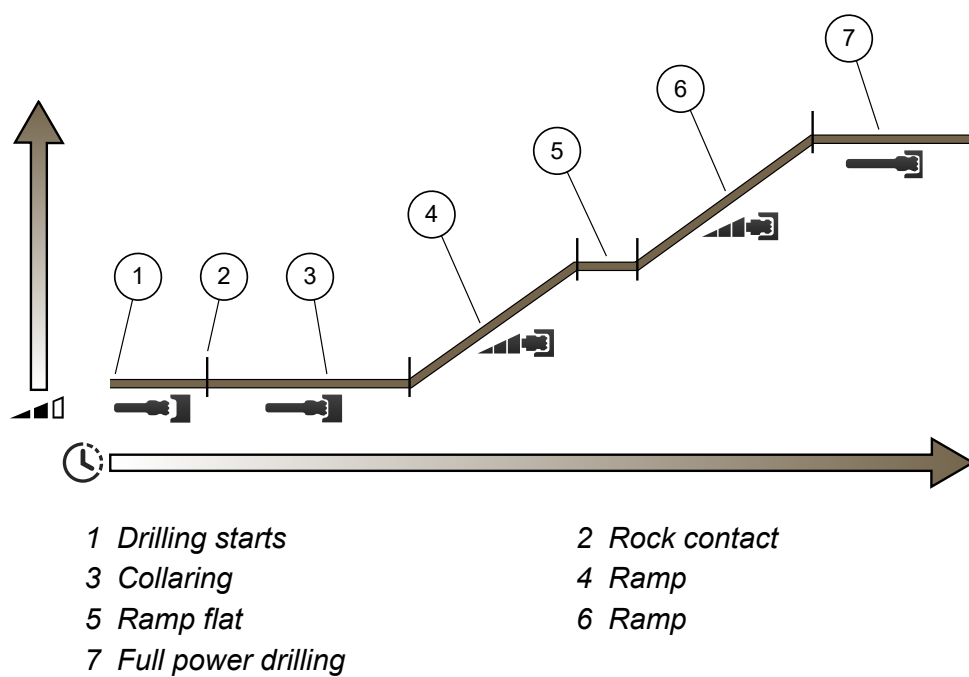


- 1 *Lifetime of the drill steels*
- 2 *Drilling accuracy*
- 3 *Penetration speed*

Note that all adjustments made are compromises between speed, accuracy, drill steel wear, overall efficiency, etc. For example, increasing penetration speed will decrease drilling accuracy and the lifetime of the drill steels. It is the operator's responsibility to prioritize when making the adjustments.



3.3 Drilling algorithm



Note!

The values shown in the following drilling adjustment images are provided for illustrative purposes only and do not necessarily correspond to actual values.

Dx4x2i Drilling adjustments

3.3.1 Rotation pressure

DD/DS/DT machines



DD/DS/DT rotation pressure

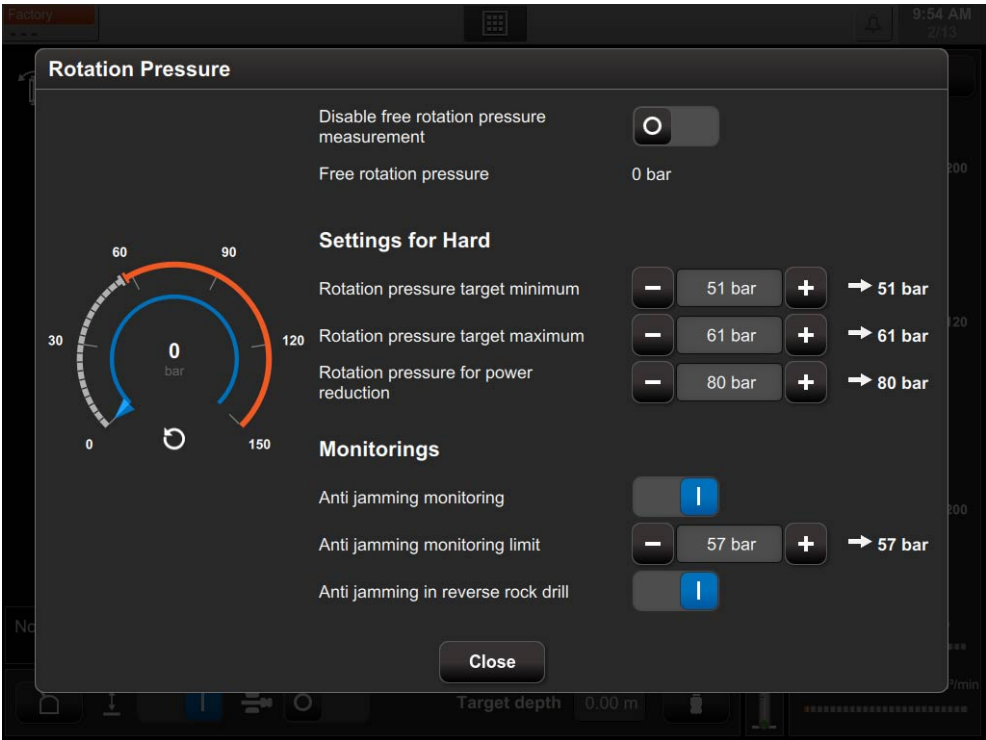
The rotation pressure is indicated as overall pressure. The optimal pressure target can be achieved during feed-percussion follow-up drilling when the best penetration rate has been reached.

DL machines

Free rotation pressure

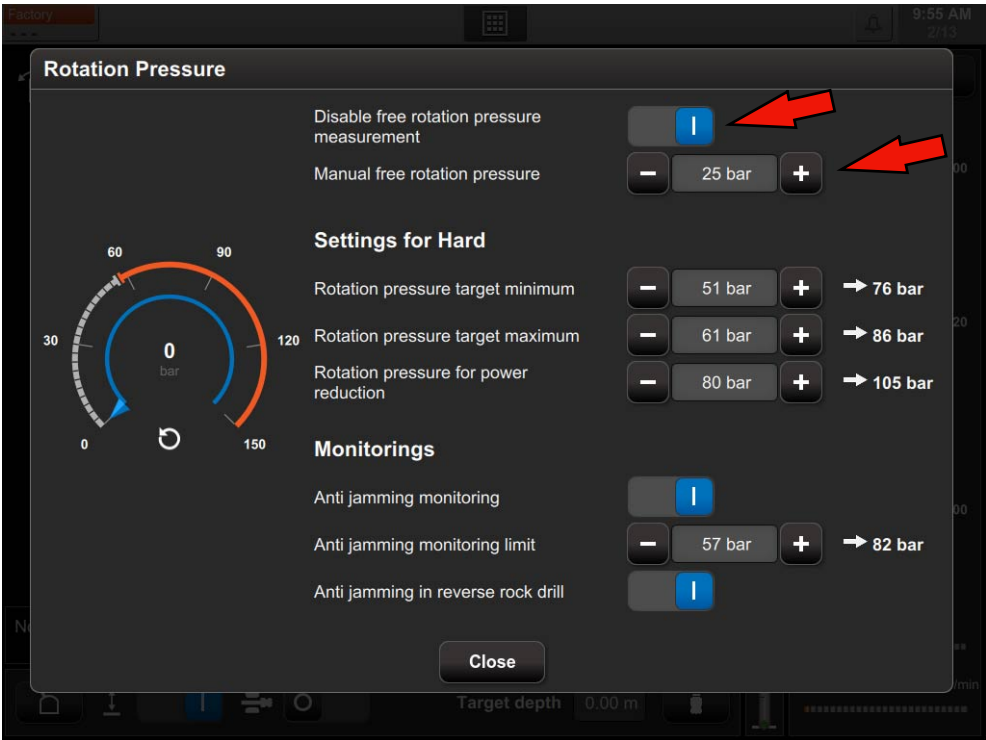
- Is used to limit the effect of rotation pressure accumulated by hole deviation or rock tools rubbing against the rock, especially in long hole drilling.
- The drill bit needs to be free of rock contact during measurement. The control system does this automatically during drilling.

Dx4x2i Drilling adjustments



DL rotation pressure, measured

The free rotation pressure is measured after every drill rod/pipe has been added. The value is added to the set values to get the overall target value.



DL rotation pressure, manual

The free rotation pressure is added manually, which is then added to the set values to get the overall target value.

Dx4x2i Drilling adjustments

3.3.2 Rock contact

DD/DS/DT machines

Drilling			
Hole Types	Drilling Settings	Drilling Monitorings	Rod Handling Settings
General Settings			
Advanced Drilling Settings			
Feed pressure rock detection limit		15 bar	15 bar
Collaring			
Collaring maximum time		20 s	
Collaring maximum distance		0.20 m	
Collaring rotation speed		100 rpm	
Collaring feed speed		1.5 m/min	
Collaring percussion pressure		100 bar	
Collaring feed pressure		55 bar	

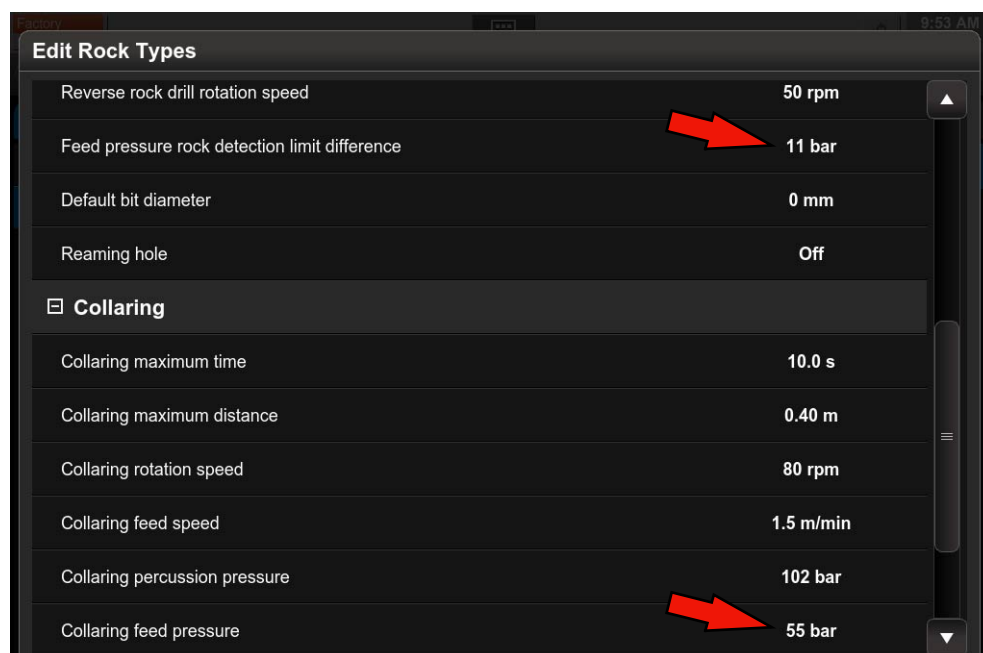
Rock contact is detected when the feed pressure rises to the rock contact threshold: *collaring feed pressure - detection limit = rock contact*.

In this example, the rock contact threshold is 40 bar. $55 \text{ bar} - 15 \text{ bar} = 40 \text{ bar}$.

After the rock detection, the collaring counters start. If the rock contact is lost, the counters will reset to zero.

Dx4x2i Drilling adjustments

DL machines



Rock contact is detected when the feed pressure rises to the rock contact threshold: collaring feed pressure - detection limit difference = rock contact.

In this example, the rock contact threshold is 44 bar. $55 \text{ bar} - 11 \text{ bar} = 44 \text{ bar}$.

After the rock detection, the collaring counters start. If the rock contact is lost, the counters will reset to zero.

3.3.3 Collaring

Collaring	
Collaring maximum time	10.0 s
Collaring maximum distance	0.20 m
Collaring rotation speed	100 rpm
Collaring feed speed	2.0 m/min
Collaring percussion pressure	120 bar
Collaring feed pressure	50 bar

Collaring parameters:

- **Collaring maximum time**
 - Defines how long, in time, collaring continues before moving to the ramp phase.
- **Collaring maximum distance**
 - Defines how long, in hole depth, collaring continues before moving to the ramp phase.
- **Collaring rotation speed**
 - Defines the rotation speed used during the collaring phase.
- **Collaring feed speed**
 - Defines the feed speed used during the collaring phase.
 - The operator can also adjust the feed speed with the feed knob. It is even possible to move the rock drill backwards.
- **Collaring percussion pressure**
 - Defines the percussion pressure used during the collaring phase.
- **Collaring feed pressure**
 - Defines the feed pressure used during the collaring phase.
 - If the value is adjusted, also pay attention to the rock contact limit value.

Longer collaring usually leads to straighter holes.

Note!

In the DD422i DC machines, there is no distance parameter for collaring. The hole depth is not measured.

Dx4x2i Drilling adjustments

3.3.4 Ramp and ramp flat

☐ Ramp and Full Power	
Ramp acceleration time	15.0 s
Ramp flat time	2.0 s

The ramp phase starts after the set collaring time or collaring distance is reached.

Ramp parameters:

- **Ramp acceleration time**
 - Defines how long the ramp acceleration time is. The control system ramps up percussion pressure, feed pressure, feed speed, and rotation speed to the desired levels.
- **Ramp flat time**
 - Removes the tension from the drilling tools.

In this example, the total ramp-up time is 17 s.

Longer ramp time usually leads to straighter holes.

3.3.5 Full power drilling

DD machines

Percussion pressure

☐ Ramp and Full Power	
Ramp acceleration time	15.0 s
Ramp flat time	2.0 s
Percussion pressure at minimum power	100 bar
Percussion pressure at maximum power	230 bar

Percussion pressure parameters:

- **Percussion pressure at minimum power**
- **Percussion pressure at maximum power**

Adjusting the drilling power using the drilling power knob changes the percussion pressure between the minimum and maximum values.

Drilling power	0...  ...100 %
Percussion pressure	100... ...230 bar

Dx4x2i Drilling adjustments

Feed pressure

Ramp and Full Power	
Ramp acceleration time	15.0 s
Ramp flat time	2.0 s
Percussion pressure at minimum power	100 bar
Percussion pressure at maximum power	220 bar
Feed pressure at minimum power	55 bar
Feed pressure at maximum power	130 bar

Feed pressure parameters:

- **Feed pressure at minimum power**
- **Feed pressure at maximum power**

Adjusting the drilling power using the drilling power knob changes the feed pressure between the minimum and maximum values.

Drilling power	0...  ...100 %
Feed pressure	55... ...130 bar

Feed speed

Advanced Drilling Settings		
Feed pressure rock detection limit	20 bar	20 bar
Maximum feed speed at minimum power	1.0 m/min	1.0 m/min
Maximum feed speed at maximum power	8.0 m/min	8.0 m/min

Feed speed parameters:

- **Maximum feed speed at minimum power**
- **Maximum feed speed at maximum power**

Adjusting the drilling power using the drilling power knob changes the feed speed maximum level between the two maximum values.

In this example, the control system is demanding 8 m/min speed from the feed direction valve at maximum drilling power setting.

Drilling power	0...  ...100 %
Feed speed	1... ...8 m/min

Dx4x2i Drilling adjustments

Rotation pressure and speed

Ramp and Full Power	
Ramp acceleration time	15.0 s
Feed pressure at maximum power	150 bar
Rotation speed during drilling	255 rpm
Torque control: Rotation pressure target at minimum power	40 bar
Torque control: Rotation pressure target at maximum power	65 bar

Rotation speed parameter:

- **Rotation speed during drilling**

Rotation pressure parameters:

- **Torque control: Rotation pressure target at minimum power**
- **Torque control: Rotation pressure target at maximum power**

Rotation speed setting is for full power drilling. Adjusting the rotation knob changes the target rotation speed when drilling with feed-percussion follow up.

In torque control mode, adjusting the rotation knob changes the rotation target pressure.

Drilling power	0...  ...100 %
Rotation pressure	40... ...65 bar

DL machines

Percussion pressure

Ramp and full power	
Ramp time	5.0 s
Maximum feed pressure	110 bar
Minimum percussion pressure	100 bar
Maximum percussion pressure	190 bar

Percussion pressure parameters:

- **Minimum percussion pressure**
- **Maximum percussion pressure**

Adjusting the drilling power using the drilling power knob changes the percussion pressure between the minimum and maximum values.

Drilling power	0...  ...100 %
Percussion pressure	100... ...190 bar

Dx4x2i Drilling adjustments

Feed pressure

☐ Ramp and full power	
Ramp time	5.0 s
Ramp flat time	1.0 s
Rotation speed during drilling	100 rpm
Minimum feed speed during drilling	3.0 m/min
Maximum feed speed during drilling	4.0 m/min
Maximum feed pressure	110 bar

Feed pressure parameter:

- **Maximum feed pressure**

Adjusting the drilling power using the drilling power knob changes the feed pressure between the minimum and maximum values.

Depending on the drilling direction, the minimum measured feed pressure can vary between 0 bar to collaring feed pressure. In this example the collaring feed pressure is 55 bar.

Drilling power	0...  ...100 %
Feed pressure	0–55... ...110 bar

Feed speed

☐ Ramp and full power	
Rotation speed during drilling	100 rpm
Minimum feed speed during drilling	3.0 m/min
Maximum feed speed during drilling	4.0 m/min

Feed speed parameters:

- **Minimum feed speed during drilling**
- **Maximum feed speed during drilling**

Adjusting the drilling power using the drilling power knob changes the feed speed between the minimum and maximum values.

Drilling power	0...  ...100 %
Feed speed	3.0... ...4.0 m/min

Rotation pressure and speed

Ramp and full power	
Ramp time	5.0 s
Ramp flat time	1.0 s
Rotation speed during drilling	100 rpm
Maximum feed pressure	150 bar
Rotation pressure target minimum	60 bar
Rotation pressure target maximum	70 bar

Rotation speed parameter:

- **Rotation speed during drilling**

Rotation pressure parameters:

- **Rotation pressure target minimum**
- **Rotation pressure target maximum**

Rotation speed setting is for full power drilling. Adjusting the rotation knob changes the target rotation speed when drilling with feed-percussion follow up.

In torque control mode, adjusting the rotation knob changes the rotation target pressure.

Drilling power	0...  ...100 %
Rotation pressure	60... ...70 bar

DS machines

Percussion pressure

Ramp and full power	
Ramp time	5.0 s
Maximum feed pressure	150 bar
Minimum percussion pressure	100 bar
Maximum percussion pressure	190 bar

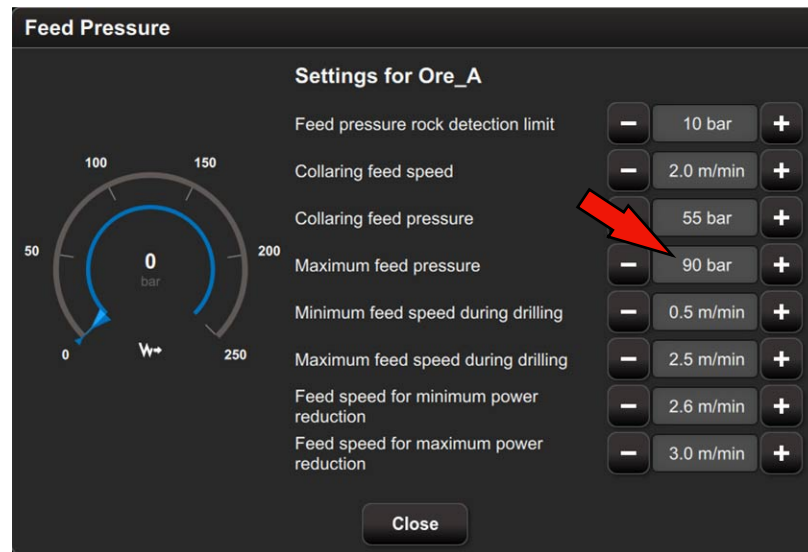
Percussion pressure parameters:

- **Minimum percussion pressure**
- **Maximum percussion pressure**

Adjusting the drilling power using the drilling power knob changes the percussion pressure between the minimum and maximum values.

Drilling power	0...  ...100 %
Percussion pressure	100... ...190 bar

Feed pressure



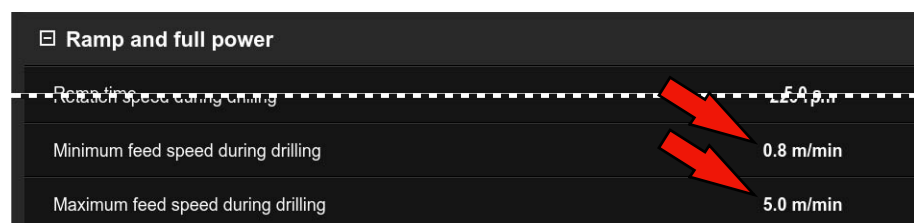
Feed pressure parameter:

- Maximum feed pressure**

Adjusting the drilling power using the drilling power knob changes the feed pressure between the minimum and maximum values. 0% is equal to the set collaring feed pressure value. In this example the collaring feed pressure is 55 bar.

Drilling power	0...  ...100 %
Feed pressure	55... ...90 bar

Feed speed



Feed speed parameters:

- Minimum feed speed during drilling**
- Maximum feed speed during drilling**

Adjusting the drilling power using the drilling power knob changes the feed speed between the minimum and maximum values.

Drilling power	0...  ...100 %
Feed speed	0.8... ...5.0 m/min

Rotation pressure and speed

Edit Rock Types

☐ Ramp and full power

Ramp time5.0 s

Ramp flat time1.0 s

Rotation speed during drilling220 rpm

Minimum feed speed during drilling0.8 m/min

Maximum percussion pressure220 bar

Rotation pressure target minimum40 bar

Rotation pressure target maximum50 bar

Rotation speed parameter:

- **Rotation speed during drilling**

Rotation pressure parameters:

- **Rotation pressure target minimum**
- **Rotation pressure target maximum**

Rotation speed setting is for full power drilling. Adjusting the rotation knob changes the target rotation speed when drilling with feed-percussion follow up.

In torque control mode, adjusting the rotation knob changes the rotation target pressure.

Drilling power	0... ...100 %
Rotation pressure	40... ...50 bar

DT machines

Percussion pressure

☐ Ramp and Full Power	
Ramp acceleration time	15.0 s
Ramp flat time	2.0 s
Percussion pressure at minimum power	100 bar
Percussion pressure at maximum power	230 bar

Percussion pressure parameters:

- **Percussion pressure at minimum power**
- **Percussion pressure at maximum power**

Adjusting the drilling power using the drilling power knob changes the percussion pressure between the minimum and maximum values.

Drilling power	0...  ...100 %
Percussion pressure	100... ...230 bar

Feed pressure

☐ Ramp and Full Power	
Ramp acceleration time	15.0 s
Ramp flat time	2.0 s
Percussion pressure at minimum power	100 bar
Percussion pressure at maximum power	220 bar
Feed pressure at minimum power	55 bar
Feed pressure at maximum power	130 bar

Feed pressure parameters:

- **Feed pressure at minimum power**
- **Feed pressure at maximum power**

Adjusting the drilling power using the drilling power knob changes the feed pressure between the minimum and maximum values.

Drilling power	0...  ...100 %
Feed pressure	55... ...130 bar

Dx4x2i Drilling adjustments

Feed speed

Advanced Drilling Settings			
Feed pressure rock detection limit		20 bar	20 bar
Maximum feed speed at minimum power		1.0 m/min	1.0 m/min
Maximum feed speed at maximum power		8.0 m/min	8.0 m/min

Feed speed parameters:

- **Maximum feed speed at minimum power**
- **Maximum feed speed at maximum power**

Adjusting the drilling power using the drilling power knob changes the feed speed maximum level between the two maximum values.

In this example, the control system is demanding 8 m/min speed from the feed direction valve at maximum drilling power setting.

Drilling power	0...  ...100 %
Feed speed	1... ...8 m/min

Rotation pressure and speed

Ramp and Full Power	
Ramp acceleration time	150 ms
Feed pressure at maximum power	150 bar
Rotation speed during drilling	255 rpm
Torque control: Rotation pressure target at minimum power	40 bar
Torque control: Rotation pressure target at maximum power	65 bar

Rotation speed parameter:

- **Rotation speed during drilling**

Rotation pressure parameters:

- **Torque control: Rotation pressure target at minimum power**
- **Torque control: Rotation pressure target at maximum power**

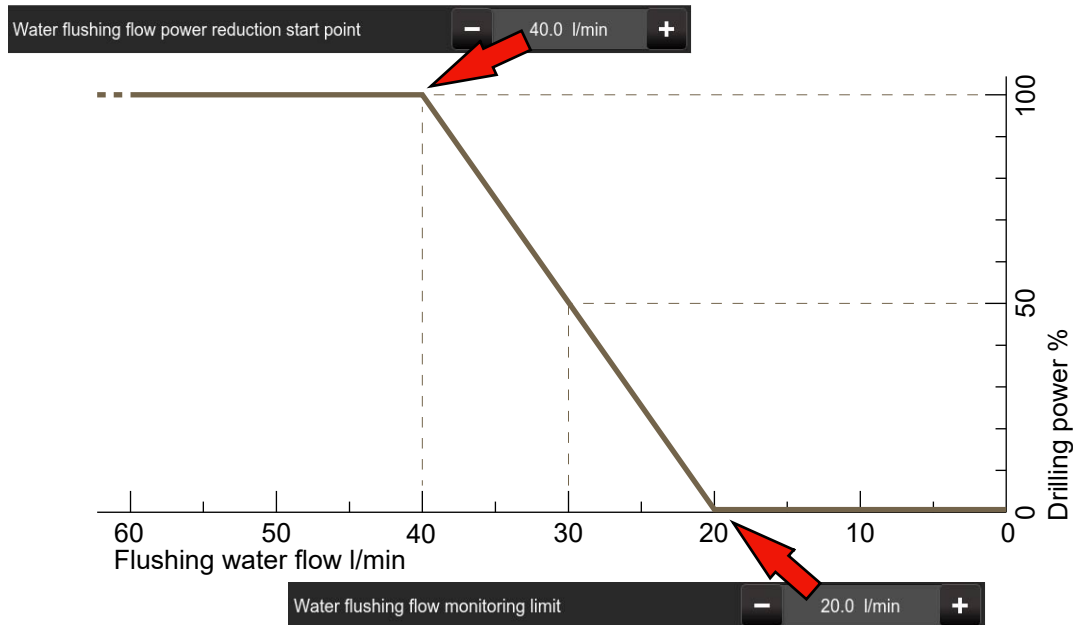
Rotation speed setting is for full power drilling. Adjusting the rotation knob changes the target rotation speed when drilling with feed-percussion follow up.

In torque control mode, adjusting the rotation knob changes the rotation target pressure.

Drilling power	0...  ...100 %
Rotation pressure	40... ...65 bar

3.3.6 Power reductions

The working principle of a power reduction



A power reduction functionality reduces drilling power from 100% to 0% according to the set values. In this example, the **Water flushing flow power reduction start point** is set to 40.0 l/min, and the **Water flushing flow monitoring limit** is set to 20.0 l/min.

The control system reduces the drilling power according to the graph. When the flushing water flow drops below 40.0 l/min, the control system starts reducing drilling power. The more the flushing flow drops, the more the control system reduces the drilling power until, at 20.0 l/min flushing water flow, it is at 0%. If the water flow starts to increase again while the power is reduced, the control system will increase the drilling power accordingly (but not over the drilling power level the operator has set).

If the operator uses a lower drilling power setting, for example 50%, the reduction of the drilling power starts only after the water flow has dropped below 30.0 l/min, even if the **Water flushing flow power reduction start point** is set to 40.0 l/min. The set start point for the power reduction is the actual start point only when the drilling power is set to 100%. If the drilling power is set to anything less, the power reduction starting point will change according to the graph.

If the operator uses a lower drilling power setting, for example 50%, and the flushing water flow is at 30.0 l/min, trying to increase the drilling power will not affect the actual drilling power as the power reduction will prevent it. Only after the flushing water flow starts to increase will the drilling power increase accordingly.

Note!

Several power reductions can affect the actual drilling power at the same time. Trying to adjust the drilling settings while any power reduction is active will result in faulty drilling settings.

Flushing flow

DD/DS/DT machines



Water flushing flow power reduction start point

The control system will start reducing drilling power when the flushing water flow drops below the set value.

Water flushing flow monitoring limit

If the flushing water flow drops below the set value, the control system will try to open the blockage by running the rock drill backwards and forwards rapidly with the percussion on.

Air flushing flow power reduction start point

The control system will start reducing drilling power when the flushing air flow drops below the set value.

Air flushing flow monitoring limit

If the flushing air flow drops below the set value, the control system will try to open the blockage by running the rock drill backwards and forwards rapidly with the percussion on.

Flushing flow, water	40... ..20 l/min
Flushing flow, air	1.5... ..1.0 m³/min
Drilling power	100...  ...0 %

DL machines

Flushing Flow

Settings for Hard

Drilling flushing method

No flushing

+

⬅️

Uncoupling flushing method

Water flushing

▼

⬅️

Monitorings



0.0 m³/min

0.00

0.01

0.02

0.03

0.04

0.05

0.06

0.07

0.08

0.09

0.10

0.11

0.12

0.13

0.14

0.15

0.16

0.17

0.18

0.19

0.20

0.21

0.22

0.23

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0.96

0.97

0.98

0.99

1.00

Flushing flow monitoring

⬅️

I

➡️

Water flow monitoring limit

−

100.0 l/min

+

Water flow power reduction limit

−

150.0 l/min

+

Air flow monitoring limit

−

5.00 m³/min

+

Air flow power reduction limit

−

6.00 m³/min

+

Close

Water flow power reduction limit

The control system will start reducing drilling power when the flushing water flow drops below the set value.

Water flow monitoring limit

If the flushing water flow drops below the set value, the control system will try to open the blockage by running the rock drill backwards and forwards rapidly with the percussion on.

Air flow power reduction limit

The control system will start reducing drilling power when the flushing air flow drops below the set value.

Air flow monitoring limit

If the flushing air flow drops below the set value, the control system will try to open the blockage by running the rock drill backwards and forwards rapidly with the percussion on.

Flushing flow, water	150...	...100 l/min
Flushing flow, air	6.0...	...5.0 m³/min
Drilling power	100...	 ...0 %

Rotation pressure

DD/DT machines

☐ Power Reduction and Anti-Jamming	
Rotation pressure for power reduction	100 bar
Rotation pressure for anti-jamming monitoring limit	120 bar
Feed speed power reduction start point	6.0 m/min

Rotation pressure for power reduction

The control system will start reducing drilling power when the rotation pressure rises above the set value.

Rotation pressure for anti-jamming monitoring limit

The control system will change the feed direction when the rotation pressure rises above the set value.

DL machines

☐ Power reductions	
Rotation pressure for power reduction	75 bar
Feed speed for minimum power reduction	3.0 m/min
Feed speed for maximum power reduction	4.0 m/min
Water flow monitoring limit	100.0 l/min
Water flow power reduction limit	150.0 l/min

Rotation pressure for power reduction

The control system will start reducing drilling power when the rotation pressure rises above the set value.

DS machines

☐ Power reductions	
Rotation pressure for power reduction	70 bar
Feed speed for minimum power reduction	2.6 m/min
Feed speed for maximum power reduction	7.0 m/min
Minimum power water flow limit	15 l/min
Water flow power reduction limit	30 l/min

Rotation pressure for power reduction

The control system will start reducing drilling power when the rotation pressure rises above the set value.

Dx4x2i Drilling adjustments

Feed speed

DD/DT machines

☐ Power Reduction and Anti-Jamming	
Rotation pressure for power reduction	100 bar
Rotation pressure for anti-jamming monitoring limit	120 bar
Feed speed power reduction start point	6.0 m/min

Feed speed power reduction start point

The control system will start reducing drilling power when the feed speed rises above the set value.

Note!

There is no feed speed reduction parameter available in the DD422i DC machines.

DL machines

☐ Power reductions	
Rotation pressure for power reduction	75 bar
Feed speed for minimum power reduction	3.0 m/min
Feed speed for maximum power reduction	4.0 m/min
Water flow monitoring limit	100.0 l/min
Water flow power reduction limit	150.0 l/min

Feed speed for minimum power reduction

The control system will start reducing drilling power when the feed speed rises above the set value.

Feed speed for maximum power reduction

The control system fully reduces drilling power when the set feed speed value is reached.

By adjusting the maximum power reduction value, you can make the power reduction more aggressive or smoother. A smaller gap between the set minimum and maximum values means more aggressive power reduction, while a larger gap between the minimum and maximum values means smoother power reduction.

DS machines

☐ Power reductions	
Rotation pressure for power reduction	70 bar
Feed speed for minimum power reduction	2.6 m/min
Feed speed for maximum power reduction	7.0 m/min
Minimum power water flow limit	15 l/min
Water flow power reduction limit	30 l/min

Feed speed for minimum power reduction

The control system will start reducing drilling power when the feed speed rises above the set value.

Feed speed for maximum power reduction

The control system fully reduces drilling power when the set feed speed value is reached.

By adjusting the maximum power reduction value, you can make the power reduction more aggressive or smoother. A smaller gap between the set minimum and maximum values means more aggressive power reduction, while a larger gap between the minimum and maximum values means smoother power reduction.

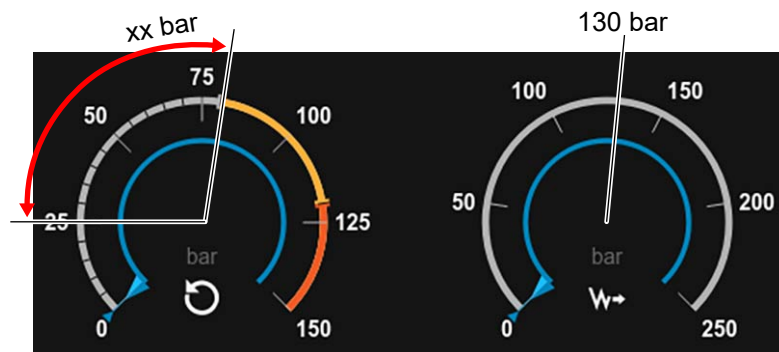
Dx4x2i Drilling adjustments

3.3.7 Feed-percussion follow-up

DD/DS/DT machines

Ramp and Full Power	
Percussion pressure at minimum power	100 bar
Percussion pressure at maximum power	230 bar
Feed pressure at minimum power	55 bar
Feed pressure at maximum power	130 bar

Advanced Drilling Settings			
Feed pressure rock detection limit	15 bar	15 bar	15 bar
Maximum feed speed at minimum power	1.0 m/min	1.0 m/min	1.0 m/min
Maximum feed speed at maximum power	8.0 m/min	8.0 m/min	8.0 m/min
Feed speed power reduction aggressiveness	Low	Low	Low



When drilling with 100% drilling power, the control system keeps the percussion pressure at the set 230 bar (**Percussion pressure at maximum power**) and feed pressure at the set 130 bar (**Feed pressure at maximum power**), if no power reduction limits are reached.

Rotation pressure can be anything between the free rotation pressure and the rotation pressure power reduction limit.

Note that the control system is always monitoring all the power reduction limits. If any power reduction limit is reached, the control system will begin to reduce the drilling power. When the drilling power is reduced, the set maximum pressure and speed values are no longer reached until the limiting power reduction is no longer active.

With the feed-percussion follow-up, the control system monitors the changes in feed pressure to adjust the percussion pressure. The percussion pressure varies in relation to the feed pressure.

Dx4x2i Drilling adjustments

3.3.8 Torque control

DD machines

Ramp and Full Power	
Pump acceleration time	20.0 s
Feed pressure at minimum power	55 bar
Feed pressure at maximum power	130 bar
Rotation speed during drilling	170 rpm
Torque control: Rotation pressure target at minimum power	40 bar
Torque control: Rotation pressure target at maximum power	65 bar

Advanced Drilling Settings	
Feed pressure lock detection limit	15 bar
Torque control: Maximum negative feed pressure deviation	40.0 bar
Torque control: Maximum positive feed pressure deviation	30.0 bar



When drilling with 100% drilling power, the control system tries to keep the rotation pressure constantly at the set 65 bar (**Torque control: Rotation pressure target at maximum power**) by adjusting the feed pressure from -40 bar (**Torque control: Maximum negative feed pressure deviation**) to +30 bar (**Torque control: Maximum positive feed pressure deviation**) from the set 130 bar (**Feed pressure at maximum power**).

In this example, the feed pressure can vary from 90 bar to 160 bar to keep the rotation pressure at 65 bar.

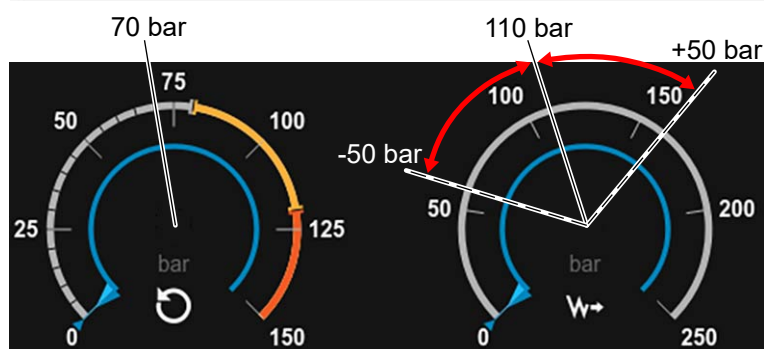
When the rotation pressure increases, the feed pressure decreases. When the rotation pressure decreases, the feed pressure increases.

Note! *The Feed-percussion follow-up is always active in the background.*

Dx4x2i Drilling adjustments

DL machines

Ramp and full power	
Maximum feed speed during running	2.5 m/min
Maximum feed pressure	110 bar
Minimum percussion pressure	100 bar
Maximum percussion pressure	190 bar
Rotation pressure target minimum	60 bar
Rotation pressure target maximum	70 bar
Torque control maximum positive feed pressure deviation	50.00 bar
Torque control maximum negative feed pressure deviation	50.00 bar



When drilling with 100% drilling power, the control system tries to keep the rotation pressure constantly at the set 70 bar (**Rotation pressure target maximum**) by adjusting the feed pressure from -50 bar (**Torque control: Maximum negative feed pressure deviation**) to +50 bar (**Torque control: Maximum positive feed pressure deviation**) from the set 110 bar (**Maximum feed pressure**).

In this example, the feed pressure can vary from 60 bar to 160 bar to keep the rotation pressure at 70 bar.

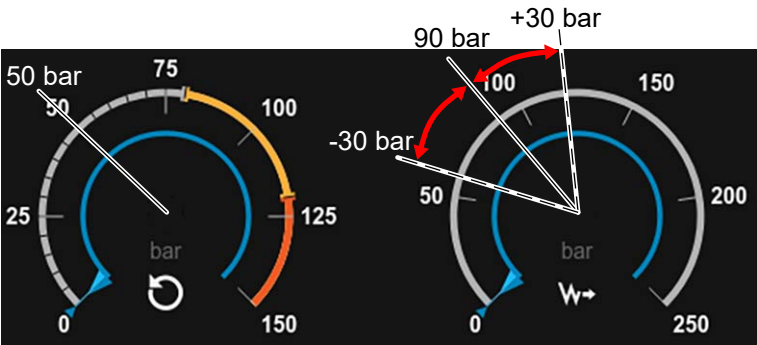
When the rotation pressure increases, the feed pressure decreases. When the rotation pressure decreases, the feed pressure increases.

Note!

The Feed-percussion follow-up is always active in the background.

DS machines

Ramp and full power	
Ramp time	5.0 min.
Maximum feed speed during drilling	1.5 m/min.
Maximum feed pressure	90 bar
Minimum percussion pressure	110 bar
Maximum percussion pressure	220 bar
Rotation pressure target minimum	40 bar
Rotation pressure target maximum	50 bar
Torque control maximum positive feed pressure deviation	30 bar
Torque control maximum negative feed pressure deviation	30 bar



When drilling with 100% drilling power, the control system tries to keep the rotation pressure constantly at the set 50 bar (**Rotation pressure target maximum**) by adjusting the feed pressure from -30 bar (**Torque control: Maximum negative feed pressure deviation**) to +30 bar (**Torque control: Maximum positive feed pressure deviation**) from the set 90 bar (**Maximum feed pressure**).

In this example, the feed pressure can vary from 60 bar to 120 bar to keep the rotation pressure at 50 bar.

When the rotation pressure increases, the feed pressure decreases. When the rotation pressure decreases, the feed pressure increases.

Note! *The Feed-percussion follow-up is always active in the background.*

4 ROCK DRILL SPECIFICATIONS

4.1 DD machines

Rock drill	Max. percussion pressure	Max. percussion frequency	Max. rotation speed
RD520			
OMS 80	220 bar	74 Hz	400 rpm
OMS 125	220 bar	74 Hz	280 rpm
OMS 160	220 bar	74 Hz	225 rpm
RD525			
OMS 80	235 bar	93 Hz	400 rpm
OMS 125	235 bar	93 Hz	280 rpm
OMS 160	235 bar	93 Hz	225 rpm
RDX5			
OMS 80	220 bar	67 Hz	365 rpm
OMS 125	220 bar	67 Hz	265 rpm
OMS 160	220 bar	67 Hz	210 rpm

Note! *The maximum specified values for a rock drill are not always the best values to use when drilling.*

Dx4x2i Drilling adjustments

4.2 DL4x2i(E) machines

Rock drill	Max. percussion pressure	Max. percussion frequency	Max. rotation speed
RD921			
OMT250	200 bar	71 Hz	250 rpm
OMT315	200 bar	71 Hz	190 rpm
RD925			
OMT250	190 bar	69 Hz	250 rpm
OMT315	190 bar	69 Hz	190 rpm
OMT400	190 bar	69 Hz	150 rpm
RD927			
OMT315	190 bar	56 Hz	190 rpm
OMT400	190 bar	56 Hz	150 rpm
OMT500	190 bar	56 Hz	120 rpm

Note! *The maximum specified values for a rock drill are not always the best values to use when drilling.*

Dx4x2i Drilling adjustments

4.3 DS412i

Rock drill	Max. percussion pressure	Max. percussion frequency	Max. rotation speed
RD314			
OMS 100	160 bar	110 Hz	530 rpm
OMS 125	160 bar	110 Hz	430 rpm

Note! *The maximum specified values for a rock drill are not always the best values to use when drilling.*

Dx4x2i Drilling adjustments

4.4 DS422i

Rock drill	Max. percussion pressure	Max. percussion frequency	Max. rotation speed
RD414			
OMS 160	220 bar	78 Hz	300 rpm
OMS 200	220 bar	78 Hz	240 rpm

Note! *The maximum specified values for a rock drill are not always the best values to use when drilling.*



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